

Syed Afraz

Machine Learning Engineer

Work Experience

CareCloud
AI Engineer 07/2025 – Present

- Led automation of healthcare RCM softwares, improving operational efficiency and reducing manual intervention.
- Designed and deployed AI-based solutions for various healthcare applications to enhance user experiences.
- Contributed to system architecture, ensuring scalability and maintainability using asynchronous programming, multiprocessing, and multi-threading techniques.

Automotive Artificial Intelligence - AAI
AI Developer 02/2025 – 06/2025

- Developed and deployed an intelligent text autocompletion system using Large Language Models (LLMs) and FastAPI
- Designed and implemented backend services, including user profile management, authentication, and session handling using FastAPI

CARE Pvt. Ltd. – Center for Advanced Research in Engineering
Software Intern 08/2024 – 10/2025

- Designed website workflows, enhancing user experience of Seller and Customer Portals.
- Designed an ERD of the website with over 30 entities to structure the Database
- Worked on AI modules, including GANs, ResNet, MLP, and Random Forest

Projects

Synthetic Image Generation and Identification of Gemstones
Final Year Project 10/2023 – 05/2024

- Built Vanilla, Conditional, and DCGAN models to generate synthetic gemstone images, achieving an FID score of 46.
- Trained a CNN for gemstone classification, reaching 88% accuracy on real and generated images.
- Developed IoT hardware with Bluetooth control to capture high-quality gemstone images for model training.

Financial News Analysis and Stock Prediction 06/2024 – 08/2024

- Investigated the relationship between financial news sentiment and stock market trends using NLP techniques.
- Performed EDA and sentiment analysis on news articles, and built an LSTM model to predict stock price movements.

AI Virtual Assistant chatbot: Jessi 11/2024 – 12/2024

- Developed a voice-responsive AI assistant using Gemini Chatbot API to automate user queries and actions.
- Integrated features such as website navigation, news updates, and music playback for enhanced user interaction.

Traceability Solution for Hierarchical Relationships
- 11/2024 – 02/2025

- Built a FastAPI backend to compute multi-level hierarchical traceability across system elements.
- Deployed the solution using Docker and Kubernetes, with MongoDB for persistent, scalable storage.

Contact

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Technical Skills

Languages: Python C++ SQL MIPS
Dart/Flutter LaTeX

Technologies: FastAPI GANs LLMs Ku-
bernetes Flask Data Structures & Algorithms

Tools: MongoDB Kaggle Docker GCP
MATLAB MySQL JIRA Blender Unity
Arduino IDE Git/GitHub Open Telemetry

libraries : TensorFlow Keras PyTorch
LangChain Scikit-learn Pandas

Education

NUST
Bachelor of Computer Engineering
2020-2024

Certifications

- Certificate of **Machine Learning Specialization** from DeepLearning.AI and Stanford.
- Kaggle certification in '**Pandas Library**'
- Certificate of KPMG- Data Analytics Consulting Virtual Internship
- Certified from LinkedIn in "**Docker for Developers By: Emmanuel Henri**"

Publication

- **Development and Industrial Implementation of Pakistan's First Integrated Biofuel Production Process: A Pathway Toward Green Energy Transition**, The China International UGS Academic Conference, October 2025.

Extra-curricular Activities

- Socials Team Lead: Managed events and activities for COMPECC & ICODT 2024.
- Director, Archery Club: Led club activities at CEME, NUST.
- IT Volunteer: Contributed to IT operations for Pakistan Youth Squad.